

JobKit - Teradata DBA interview questions

212 questions for Teradata DBA. Single-file download.

1. What is Teradata?

A shared-nothing MPP database for large analytic workloads.

2. What is an AMP?

Access Module Processor: a parallel worker that stores and processes part of the data.

3. What is a PE?

Parsing Engine: parses SQL, optimizes, and sends steps to AMPs.

4. What is BYNET?

The interconnect that moves messages and data between PEs and AMPs.

5. What is a vproc?

A virtual processor such as AMP or PE running on a node.

6. What is a clique?

A set of nodes that share disks so a failed node can fail over.

7. What is a hot standby node?

A spare node that can take over vprocs after a failure.

8. What is Fallback?

A second copy of a table on another AMP for availability.

9. When do you use Fallback?

Critical tables that must survive AMP or disk loss.

10. What is a Primary Index?

The distribution key that decides which AMP stores a row.

11. Unique vs non-unique PI?

UPI enforces uniqueness; NUPI allows duplicates and can skew.

12. What is a Secondary Index?

An extra access path; it has a subtable and extra I/O.

13. What is a Join Index?

A stored join or aggregation that the optimizer can use.

14. What is PPI?

Partitioned Primary Index: partitions rows inside each AMP.

15. What is a NoPI table?

A table without a PI; used for fast staging loads.

16. What is a sparse map?

A map that uses a subset of AMPs for smaller tables.

17. What is a contiguous map?

The default map that uses all AMPs in the system.

18. What is hashing in Teradata?

The PI value is hashed to pick the AMP.

19. What is skew?

Uneven row or CPU distribution across AMPs.

20. How do you find skewed tables?

Compare AMP row counts or use DBQL and Viewpoint.

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21. What is a hot AMP?

An AMP doing more work than others, often from skew.

22. What is spool space?

Temporary space for intermediate query results.

23. What is perm space?

Permanent table storage quota.

24. What is temp space?

Space for global temporary tables.

25. User vs database in Teradata?

A user can log on; a database is a space container.

26. What is a profile in Teradata?

A set of account, spool, and password settings for users.

27. What is a role?

A bundle of rights you grant to users.

28. What is GRANT vs role?

GRANT is a privilege; a role groups many privileges.

29. What is a macro?

A stored SQL snippet executed with EXEC.

30. What is a stored procedure?

Procedural SQL with logic, not just a macro.

31. What is a volatile table?

Session-scoped table that disappears on logoff.

32. Global temporary vs volatile?

GTT definition is permanent; data is session-scoped.

33. What is a derived table?

An inline subquery in the FROM clause.

34. What is a CTE?

WITH clause query that names a result set.

35. What is COLLECT STATISTICS?

Gather data demographics so the optimizer can plan well.

36. When do you collect stats?

After large loads, skew, or bad plans; not blindly every hour.

37. What is SUMMARY stats?

A cheaper stats option on some releases for quick demographics.

38. What is sampled stats?

Stats on a sample; faster but less accurate.

39. How do you see a plan?

EXPLAIN SELECT ...

40. What is a confidence level in EXPLAIN?

How sure the optimizer is about row estimates.

41. What is a product join?

Every row compared to every row; often a warning sign.

42. What is a merge join?

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Join on sorted/hashed keys; usually cheaper.

43. What is a hash join?

Build a hash table of the smaller set and probe it.

44. What is a nested join?

Index-driven lookup join.

45. What is redistribution?

Moving rows across AMPs so join keys co-locate.

46. What is duplication of a table in a join?

Copying a small table to all AMPs.

47. What is a residual condition?

A filter applied after the join or retrieve step.

48. What is a lock in Teradata?

Access, read, write, or exclusive lock on a table or row hash.

49. What is a deadlock?

Two sessions waiting on each other; one is aborted.

50. How do you reduce lock contention?

Shorter transactions, nightly batch windows, and row-hash locks.

51. What is a transaction in Teradata?

BT/ET or ANSI transaction around SQL.

52. BTET vs ANSI mode?

BTET is Teradata transaction mode; ANSI is commit-oriented.

53. What is FastLoad?

Utility to load empty tables quickly with two phases.

54. FastLoad limitation?

Target must be empty; no SI during load in classic use.

55. What is MultiLoad?

Utility for insert/update/delete on populated tables.

56. What is TPump?

Low-volume continuous load with row-hash locks.

57. What is FastExport?

High-volume export from Teradata.

58. What is BTEQ?

CLI for SQL, scripts, and simple imports.

59. What is TPT?

Teradata Parallel Transporter: modern load/export framework.

60. TPT vs legacy utilities?

TPT can replace FastLoad/MultiLoad/FastExport with operators.

61. What is a load slot?

A limit on concurrent load jobs.

62. What is a checkpoint in MultiLoad?

A restart point after a failure.

63. What is an error table?

Where utilities land rejected rows.

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64. What is a UV table?

Uniqueness violation table in MultiLoad.

65. What is Viewpoint?

Web console for health, queries, and alerts.

66. What is QueryGrid?

Fabric to query Hadoop, Oracle, and other systems from Teradata.

67. What is TASM?

Teradata Active System Management: workload rules and throttles.

68. What is a workload in TASM?

A class of queries with priority and limits.

69. What is a throttle?

A cap on concurrency for a workload.

70. What is a filter in TASM?

A rule that rejects or delays some queries.

71. What is exception handling in TASM?

Move or abort queries that exceed CPU or skew.

72. What is WD-Default?

The default workload when nothing else matches.

73. What is a utility session?

Sessions used by load/export utilities.

74. What is BAR?

Backup and restore, often with DSA or NetBackup.

75. What is DSA?

Data Stream Architecture for backups.

76. Full vs incremental backup?

Full copies everything; incremental copies changes.

77. How do you restore one table?

Use BAR object-level restore if the backup supports it.

78. What is a dump vs backup?

Dump often means a system dump for support; backup is data protection.

79. What is PDBA vs production DBA?

PDBA may own platform; production DBA owns day-to-day jobs and users.

80. What is space reclamation?

Freeing perm after deletes; packing and fallback copies matter.

81. What is a cylinder?

A unit of disk allocation on an AMP.

82. What is a table header?

Metadata stored with the table on each AMP.

83. What is a journal?

Before/after image used for recovery on some tables.

84. Permanent journal vs transient?

Permanent is extra protection; transient is for transactions.

85. What is TVS?

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Teradata Virtual Storage: storage virtualization layer.

86. What is FSYS?

File system layer under AMPs.

87. What is a node crash recovery?

Vprocs migrate; cliques and HSN determine downtime.

88. What is a database restart?

TPA reset: sessions drop; need to reconnect.

89. What is TPA?

Teradata parallel database process stack.

90. What is PDE?

Parallel Database Extensions: OS layer for vprocs.

91. What is a gateway?

Connection manager for client sessions.

92. What is a session?

A logged-on PE connection from a client.

93. Default session mode?

Depends on client; know if you are ANSI or Teradata mode.

94. What is a request vs a transaction?

A request is one SQL; a transaction may be many requests.

95. How do you kill a bad query?

Abort in Viewpoint or with a console abort after checking impact.

96. How do you find who filled the spool?

DBQL plus user spool limits.

97. What is DBQL?

Database Query Log: history of queries and steps.

98. What is ResUsage?

Resource usage macros for CPU, disk, and BYNET.

99. What is a heart beat in Viewpoint?

System health pulse; red means investigate now.

100. What is a blocked session?

Waiting on a lock held by another session.

101. How do you see blockers?

Lock viewer in Viewpoint or lock query views.

102. What is a deadlock timeout?

How long Teradata waits before aborting a deadlock victim.

103. What is a PPI advantage?

Partition elimination so fewer cylinders are read.

104. What is a character PPI?

Partitioning on a character column; less common than date PPI.

105. What is multilevel PPI?

More than one partition expression.

106. What is a column partition (CP)?

Columnar storage for analytic tables on some releases.

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107. What is Hybrid Row/Column?

Mix of row and column partitions.

108. What is compression in Teradata?

MVC, UDT, algorithmic, or block-level depending on release.

109. What is MVC?

Multi-Value Compression: dictionary compression on a column.

110. What is a UDT?

User-defined type.

111. What is a UDF?

User-defined function.

112. What is a table operator?

SQL-HMR style operator for custom processing.

113. What is Native Object Store (NOS)?

Read/write object storage (S3-style) from Teradata Vantage.

114. What is Vantage?

Teradata's analytics platform branding for newer releases.

115. What is QueryBand?

Name/value tags on a session or transaction for workload and audit.

116. What is an account string?

Account used for chargeback and TASM mapping.

117. What is a performance group?

Older priority scheme; TASM workloads replaced much of this.

118. What is a resource partition?

A slice of CPU reserved for a set of workloads.

119. What is tactical vs decision support?

Tactical is short; DSS is heavy reporting.

120. How do you protect tactical queries?

TASM: high priority, low throttle, and exception on long CPU.

121. What is a delayed query?

Queued by a throttle until a slot is free.

122. What is a reject in TASM?

The query is not allowed to run.

123. What is an object lock vs row hash lock?

Object is table-level; row hash is finer.

124. What is a dummy row lock?

Locking a hash with no row; still blocks that hash.

125. How do you load a large fact table daily?

Stage NoPI or FastLoad, then INSERT-SELECT with collect stats.

126. What is an INSERT-SELECT?

Load from one table into another set-based.

127. What is MERGE INTO?

Upsert based on a match condition.

128. What is UPDATE from a join?

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Set-based update using another table.

129. What is a SET vs MULTiset table?

SET rejects duplicate rows; MULTiset allows them.

130. Duplicate row check cost?

SET tables can be expensive on NUPI during inserts.

131. What is a queue table?

A table with consume semantics for messaging-style pulls.

132. What is an identity column?

System-generated numbers; be careful with distribution.

133. What is a date PI problem?

Low cardinality dates skew if used as PI.

134. Good PI for a fact table?

A high-cardinality join key, often a customer or transaction id.

135. What is referential integrity in Teradata?

Optional; soft RI is often used for optimizer hints.

136. What is a batch vs tactical window?

Batch at night with loads; tactical during business hours.

137. How do you handle a failed FastLoad?

Drop or empty the target, fix the file, restart; check error tables.

138. What is a two-phase FastLoad?

Phase 1 loading to AMPs; phase 2 sorting into the table.

139. Can FastLoad load a PI with SI?

Classic FastLoad wants no secondary indexes on the target.

140. What is a MiniBatch?

A pattern of small MultiLoad or TPT jobs.

141. What is streaming ingest?

TPump or TPT Stream for near-real-time.

142. How do you grant access to a new analyst?

Role-based: create user, grant role, set spool and profile.

143. How do you revoke a leaver?

Lock the user, revoke roles, drop or keep for audit per policy.

144. What is a password lockout?

Profile setting after failed logons.

145. What is LDAP vs TD database auth?

Enterprise directory vs Teradata passwords.

146. What is SSO to Viewpoint?

Viewpoint can use corporate identity; still map TD users.

147. What is a zone in Viewpoint?

A filtered view of systems for a team.

148. What is a portlet?

A Viewpoint dashboard widget.

149. What is a system heartbeat alert?

Pager-worthy if the database is down.

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150. How do you plan an upgrade?

Read the release, test BAR, check views, schedule an outage window.

151. What is a compat view?

Older dictionary views kept for scripts.

152. DBC vs SYSLIB?

DBC is the data dictionary; SYSLIB holds functions.

153. What is DBC.TablesV?

Dictionary view of tables.

154. How do you find table size?

SUM of currentperm from DBC.TableSizeV grouped by table.

155. How do you find user space?

DBC.DiskSpaceV or Viewpoint space portlet.

156. What is currentperm vs peakperm?

Current used vs high-water used.

157. What is a stall?

AMPs waiting on I/O or each other; check ResUsage.

158. What is a bottleneck on BYNET?

Heavy redistribution; redesign PI or join.

159. What is a CPU-bound query?

High AMP CPU; check product joins and functions on columns.

160. What is a missing stats symptom?

Wrong join type and huge spool.

161. How often to collect stats on a large fact?

After major loads; use THRESHOLD where available.

162. What is stats THRESHOLD?

Skip collect if data has not changed enough.

163. What is a copy job in BAR?

Copy backup data to another system.

164. What is a restore to a different system?

Needs compatible maps and enough AMPs or a remap strategy.

165. What is a map change?

Redistribute tables after adding AMPs.

166. What is a reconfig?

System change such as adding nodes; planned outage.

167. What is HSN vs clique failover?

HSN is a spare node; clique shares disks among nodes.

168. What is a down AMP?

AMP not processing; fallback or HSN may cover.

169. What is a catchup?

Recovering fallback copies after a component returns.

170. What is a checktable?

Utility to check table integrity.

171. What is a scandisk?

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Lower-level disk integrity check.

172. What is a perm cache?

Memory cache of perm data; watch hit rates.

173. What is FSG cache?

File segment cache for I/O.

174. What is a TVS temperature?

Hot/cold data placement on mixed storage.

175. What is a 1-AMP operation?

A query that hits a single AMP via unique PI access.

176. What is an all-AMP retrieve?

Every AMP reads its piece of the table.

177. What is a group AMP operation?

A subset of AMPs, for example sparse maps.

178. What is a join index maintenance cost?

Extra work on every base-table change.

179. When is a JI worth it?

Stable, repeated joins that dominate CPU.

180. What is an aggregate JI?

Pre-aggregated stored result.

181. What is a single-table JI?

A stored projection or extra access path on one table.

182. What is a covering index?

An index that satisfies a query without the base table.

183. What is a value-ordered NUSI?

NUSI stored in value order for range access.

184. What is a hash-ordered NUSI?

Default NUSI organization.

185. What is a unique secondary index?

Enforces uniqueness and supports lookups.

186. Cost of USI on load?

Load utilities may require dropping USI first.

187. What is a soft RI?

Optimizer uses the relationship without enforcing it.

188. What is a batch error table?

Landing for failed rows in a batch.

189. How do you handle late-arriving dimensions?

Staging and MERGE; do not invent keys.

190. What is slowly changing dimension?

Type 1 overwrite vs Type 2 history rows.

191. How does a DBA support SCD2?

PI, PPI on dates, and space for history.

192. What is a surrogate key?

Internal integer key; pick a PI that does not skew.

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193. What is a natural key?

Business key such as account number.

194. What is 3NF vs star schema?

3NF is normalized EDW; star is dimensional mart.

195. Where does Teradata sit in a lakehouse?

Often the governed EDW; NOS can query the lake.

196. What is data lineage?

Where a column came from; DBA supports via views and docs.

197. What is a view vs a table?

View is stored SQL; table is stored rows.

198. What is a recursive view?

View that refers to itself; use carefully.

199. What is a locking view?

A view that includes LOCKING modifiers.

200. LOCKING ROW FOR ACCESS?

Allows dirty reads to reduce blocking.

201. When is ACCESS locking dangerous?

When the reader must see committed consistent numbers.

202. What is a deadlock victim?

The session Teradata aborts to break a deadlock.

203. How do you tune a product join?

Stats, better join keys, or rewrite to merge/hash.

204. What is a character set issue?

Latin vs Unicode; conversions can cost CPU.

205. What is a case-specific compare?

CS vs CASESPECIFIC in comparisons.

206. What is a format in BTEQ?

How numbers and dates print in reports.

207. What is .IF ERRORCODE in BTEQ?

Branch on failure in a script.

208. What is a return code in jobs?

Scheduler uses it to fail the job.

209. How do you alert on a failed load?

Scheduler plus Viewpoint alerts plus mail.

210. What is a runbook?

Step-by-step for a repeat incident.

211. What is a CAB?

Change advisory board for production changes.

212. What is a back-out plan?

How you undo a change if it fails.